Statistical annex: presidents and the Philippine economy

16 October 2026

This annex collects every table generated by the estimation, as rendered from `output/tables/`. The full machine-readable results, including the one-shock models and the robustness checks for every administration, are the csv files in `output/tables/` listed at the end.

A. Validation

Table 1. Replication of Blinder and Watson (2016) on US data, 1949Q2–2013Q1

Criterion	Statistic	Paper	This study	Pass range	Pass
G1	D mean growth	4.35	4.346	[4.25, 4.45]	yes
G1	R mean growth	2.54	2.543	[2.44, 2.64]	yes
G1	D–R gap	1.80	1.803	[1.7, 1.9]	yes
G2	Newey–West (6) s.e. of gap	0.64	0.637	[0.54, 0.74]	yes
G2	randomization p-value (16 blocks)	0.01	0.011	[0.0, 0.03]	yes
G3	oil-explained gap, adopted reading: lags 0–6	0.50	0.481	[0.4, 0.6]	yes
G4	corr. of term oil effects with Table 7	≥ 0.90	0.996	[0.9, 1.0]	yes
G4	terms with the same sign (of 16)	≥ 14	16	[14, 16]	yes

Notes: Pass ranges were fixed before the gate was run. Growth is quarterly real GDP growth at a compounded annual rate, ALFRED vintage of 1 July 2014.

Source: Blinder and Watson (2014, Tables 1, 6 and 7); author's calculations from ALFRED and BLS data.

B. The facts (RQ1)

Table 2. Administration premiums in quarterly real GDP growth, 1981Q2–2026Q2

Administration	Quarters charged (n)	Mean growth	Premium (s.e.)	Holm p	Per-person premium (s.e.)
Marcos Sr. (1981–86)	1981Q2–1986Q2 (21)	−0.20	−4.22* (2.26)	0.44	−5.03** (2.30)
C. Aquino	1986Q3–1992Q3 (25)	3.25	−0.77 (1.38)	1.00	−1.35 (1.37)
Ramos	1992Q4–1998Q3 (24)	3.78	−0.24 (0.93)	1.00	−0.60 (0.98)
Estrada	1998Q4–2001Q1 (10)	3.48	−0.54 (0.83)	1.00	−1.03 (0.86)
Arroyo	2001Q2–2010Q3 (38)	5.09	+1.07 (0.70)	0.76	+1.13 (0.76)
Aquino III	2010Q4–2016Q3 (24)	6.24	+2.22*** (0.72)	0.02	+2.43*** (0.79)
Duterte	2016Q4–2022Q3 (24)	4.72	+0.70 (1.84)	1.00	+1.54 (1.87)
Marcos Jr.	2022Q4–2026Q2 (15)	4.59	+0.56 (0.90)	1.00	+1.99** (0.88)

Notes: Growth at a compounded annual rate, percent; premium = administration mean minus the sample mean (4.02%). Newey–West standard errors (6 lags) in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01 (unadjusted). Holm p corrects for the eight tests. Circular-shift test of no differences: p = 0.44 (raw), 0.28 (per person).

Source: Author's calculations from Philippine Statistics Authority, World Bank, FRED/ALFRED, NOAA and Federal Reserve Board data.

Table 3. Annual premiums, raw and relative to regional peers, 1962–2025

Administration	Years	Premium, PSA growth	Premium relative to IDN–MYS–THA	Relative to IDN–MYS–THA–VNM
Macapagal	4	+0.86	+1.25	
Marcos Sr.	20	−0.46	−1.73	−8.25
C. Aquino	7	−0.84	−3.26	−2.82
Ramos	6	−0.63	+0.40	−0.71
Estrada	2	−0.37	+0.14	−0.21
Arroyo	10	+0.55	+1.23	+0.64
Aquino III	6	+1.87	+2.90	+2.33
Duterte	6	−0.55	+2.16	+1.26
Marcos Jr.	3	+0.92	+2.37	+1.57

Notes: Percentage points. Years charged to the administration in office for most of the year after a one-quarter lag. Dispersion of premiums: 0.82 (raw), 1.99 (relative). Shift-test p: 1.00 (raw), 0.77 (relative). Vietnam data start in 1985, so the last column covers only 1985 for Marcos Sr.

Source: Author's calculations from Philippine Statistics Authority, World Bank, FRED/ALFRED, NOAA and Federal Reserve Board data.

C. Luck (RQ2)

Table 4. How much of each premium is luck? Joint shock model, 1983Q4–2026Q2

Administration	Premium	Explained (s.e.)	Unexplained	Oil	US demand	El Niño	Financial
Marcos Sr. (1983–86)	−6.97	−2.03 (1.63)	−4.95	+0.59	−2.67	+0.04	+0.00
C. Aquino	−0.85	−0.22 (0.48)	−0.62	−0.06	+0.32	−0.13	−0.36
Ramos	−0.32	+0.29 (0.24)	−0.61	+0.28	−0.45	−0.10	+0.56
Estrada	−0.61	−1.90 (1.17)	+1.29	+0.32	−1.03	−0.05	−1.15
Arroyo	+1.00	−0.02 (0.62)	+1.02	−0.76	+0.90	+0.18	−0.35
Aquino III	+2.14	+1.04 (0.78)	+1.10	+0.54	+0.32	−0.01	+0.19
Duterte	+0.63	+0.45 (0.29)	+0.17	+0.11	+0.01	−0.07	+0.40
Marcos Jr.	+0.49	+0.33 (0.24)	+0.16	−0.13	+0.03	+0.04	+0.40

Notes: Percentage points. Quarters as in Table 2 from 1983Q4 (Marcos Sr.: 11). Explained = the administration's average of the fitted shock component minus its sample average; the last four columns split it by shock. Each shock enters with lags 0–6. Share of the dispersion of premiums explained, S = 0.48. Wald tests that a shock's lags are all zero: oil p = 0.09, US demand p = 0.53, El Niño p = 0.84, financial conditions p = 0.005.

Source: Author's calculations from Philippine Statistics Authority, World Bank, FRED/ALFRED, NOAA and Federal Reserve Board data.

Joint model: 171 quarters, $R^2 = 0.19$. Sums of the lag coefficients: oil −0.33, US demand −1.01, El Niño −1.53, financial conditions −9.40. Wald test that all shock coefficients are zero: $p < 0.001$.

D. Inheritance (RQ3) and the same point in the term (RQ4)

Table 5. Was first-year growth inherited?

Administration	Forecast made in	Actual first-year growth	AR forecast	Bounce-back forecast	Inherited (H3 rule)
C. Aquino	1986Q2	4.51	4.31	4.64	yes
Ramos	1992Q3	2.34	3.01	1.05	no
Estrada	1998Q3	3.53	3.01	2.40	no
Arroyo	2001Q1	3.78	3.34	2.91	no
Aquino III	2010Q3	3.16	4.24	5.02	no
Duterte	2016Q3	7.14	4.24	4.81	no
Marcos Jr.	2022Q3	5.75	4.20	4.88	yes

Notes: Growth over the first four charged quarters, 100·ln(GDP_s+3 / GDP_s−1), percent. Sample mean 3.70%. 'Inherited' if the bounce-back forecast exceeds the mean by at least half of the actual excess.

Source: Author's calculations from Philippine Statistics Authority, World Bank, FRED/ALFRED, NOAA and Federal Reserve Board data.

Table 6. The same point in the term: the first 15 quarters

Administration	First 15 quarters	Raw mean	Difference (s.e.)	Luck-adjusted mean	Difference (s.e.)
C. Aquino	1986Q3–1990Q1	5.17	−0.58 (1.48)	5.14	−0.88 (2.16)
Ramos	1992Q4–1996Q2	4.54	+0.05 (0.90)	3.64	+0.62 (1.38)
Arroyo	2001Q2–2004Q4	5.09	−0.50 (0.96)	4.60	−0.35 (1.41)
Aquino III	2010Q4–2014Q2	6.02	−1.43 (1.01)	4.29	−0.04 (1.42)
Duterte	2016Q4–2020Q2	1.24	+3.35 (4.79)	1.21	+3.05 (4.54)
Marcos Jr.	2022Q4–2026Q2	4.59		4.26	

Notes: Percent, annual rate. Difference = Marcos Jr. minus the administration in the row. Luck-adjusted growth removes the fitted shock component of the joint model. Estrada (10 quarters) and Marcos Sr. (first quarters outside the sample) are not compared. Newey–West standard errors; the luck-adjusted ones ignore the estimation error of the shock component.

Source: Author's calculations from Philippine Statistics Authority, World Bank, FRED/ALFRED, NOAA and Federal Reserve Board data.

E. Robustness

Table 7. Robustness checks

Specification	Shift-test p	Luck share S	Aquino III premium	Marcos Jr. premium
Main	0.44	0.48	+2.22	+0.56
Timing lag 0	0.27	0.33	+2.15	+0.79
Timing lag 2	0.34	0.40	+2.32	+0.35
Timing lag 4	0.26	0.39	+2.77	+0.83
Excluding 2020Q2–2021Q1	0.42	0.47	+2.15	+0.50
Excluding 1984Q1–1985Q4	0.86	0.12	+1.83	+0.18
Log growth	0.43	0.43	+2.35	+0.77
Lags 1–6	0.44	0.51	+2.22	+0.56
Foreign demand US-JP-EA (from 1997)	0.44	0.11	+2.22	+0.56

Notes: All checks were specified before estimation. Premiums are raw quarterly premiums (percentage points).

Source: Author's calculations from Philippine Statistics Authority, World Bank, FRED/ALFRED, NOAA and Federal Reserve Board data.

F. Scorecard (Objective 1)

Table 8. Scorecard, annual averages 1962–2025 (z-score in parentheses)

Administration	GDP growth	Growth per person	Relative to peers	Inflation	Unemployment (1991–)
Macapagal	5.02 (+0.26)	1.92 (−0.04)	0.02 (+0.37)	5.54 (−0.34)	
Marcos Sr.	3.70 (−0.14)	1.16 (−0.26)	−2.97 (−0.51)	14.32 (+0.74)	
C. Aquino	3.32 (−0.26)	0.81 (−0.36)	−4.50 (−0.95)	10.20 (+0.23)	3.79 (+0.79)
Ramos	3.53 (−0.19)	1.24 (−0.24)	−0.84 (+0.12)	7.71 (−0.07)	3.70 (+0.64)
Estrada	3.79 (−0.11)	1.33 (−0.21)	−1.09 (+0.04)	4.96 (−0.41)	3.76 (+0.74)
Arroyo	4.71 (+0.17)	2.85 (+0.24)	−0.00 (+0.36)	4.64 (−0.45)	3.69 (+0.62)
Aquino III	6.03 (+0.57)	4.43 (+0.70)	1.67 (+0.85)	2.64 (−0.70)	3.33 (+0.00)
Duterte	3.61 (−0.17)	2.74 (+0.20)	0.93 (+0.63)	3.78 (−0.56)	2.61 (−1.23)
Marcos Jr.	5.08 (+0.28)	4.36 (+0.68)	1.14 (+0.69)	3.62 (−0.58)	2.21 (−1.90)

Notes: Percent; z = (administration mean − 1962–2025 mean) / standard deviation of the annual values. For inflation and unemployment a negative z means a lower rate than usual. No composite index is formed and administrations are not ranked. Unemployment is the ILO modelled estimate (1991–).

Source: Author's calculations from Philippine Statistics Authority, World Bank, FRED/ALFRED, NOAA and Federal Reserve Board data.

G. Files

`replication_gate.csv`, `replication_term_effects.csv`, `sa_check.csv`, `admin_calendar.csv`, `rq1_premiums_q.csv`, `rq1_premiums_a.csv`, `rq1_tests.csv`, `rq2_single.csv`, `rq2_joint.csv`, `rq2_summary.csv`, `rq3_inheritance.csv`, `rq4_same_point.csv`, `scorecard.csv`, `robustness.csv`, `paper_numbers.json`.